

Q1.

Prove the following:

$$\overline{X} + XY = \overline{X} + Y$$

Q2.

Find the logic circuit for:

$$\overline{A}BC + A\overline{B}C + \overline{A}B + ABC$$

Q:

Simplify the logic function and find the logic circuit:

$$f_{(xyz)} = M_0 + M_1 + M_2 M_3 + M_4 M_6$$

Find the reduced form for the logic function
Presented in fig :

	$\bar{Z}\bar{V}$	$\bar{Z}V$	ZV	$Z\bar{V}$
$\bar{X}\bar{Y}$		1		
$\bar{X}Y$	1	1	1	1
XY	1	1	1	1
$X\bar{Y}$		1	1	

Q.

Design the combination circuit with 3 input and 1 output if the sum of inputs equal 2 and the output equal 2.